Semiconducters (UNIT 1)

Conducter > Semicounter > Insultator

Conducter: $10^4 - 10^7$; Semicounter: $10^{-6} - 10^3$; Insultator: $10^{-4} - 10^{-6}$

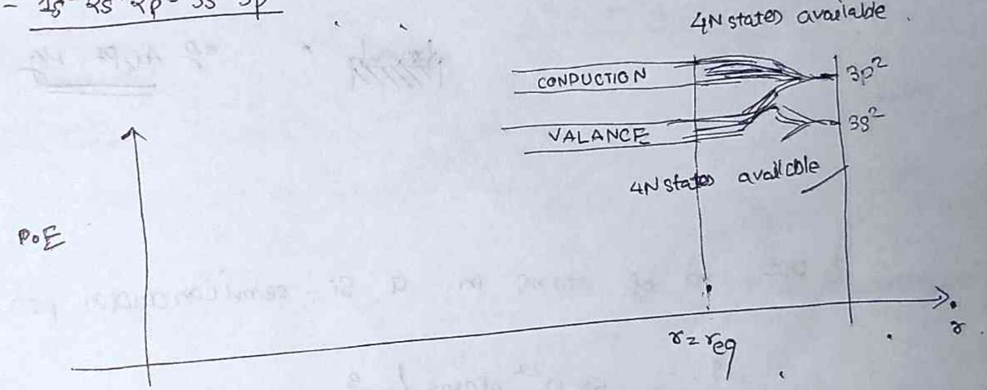
Ω^{-1} = Siemens

Units :- $S\ cm^{-1}$, $\mho\ cm^{-1}$, $mho\ cm^{-1}$

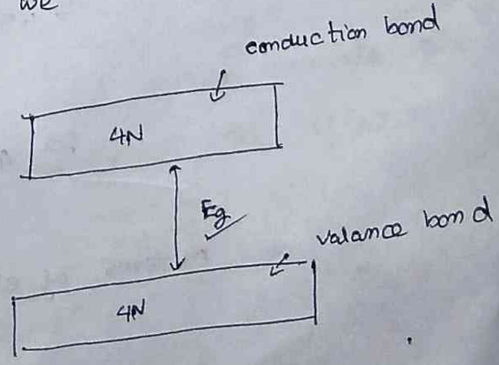
Doping as a regulator in conductivity.

-Si-O-Si
Si-O-Si

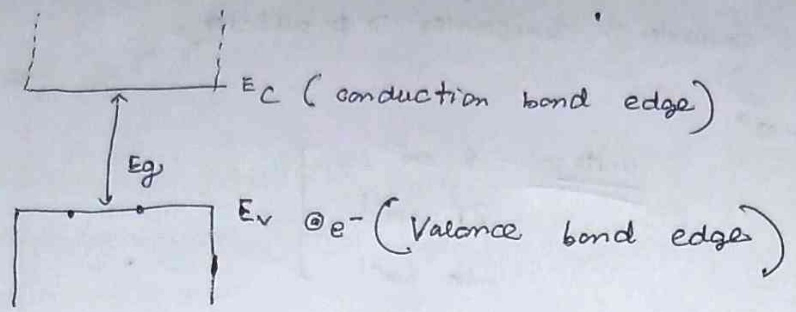
$Si_{14} = \underline{1s^2 2s^2 2p^6 3s^2 3p^2}$



Let's consider 'N' number of atoms are brought to each other, then we obtain the following fig



classification of elements on the basis of energy band gap



(i) If $E_g < 3eV$ then its a semiconducters . e.g Si, Ge, Gen etc

(ii) If $E_g > 3eV$, then its a insulator. quatz, wood

(iii) overlapping bands are found in counductors. $[E_g \approx 0]$

e.g Au, Pt, Hg

→ The no of atoms in a Si - semiconductor per unit volume is 5×10^{22} atoms$/cm^3$.

Let the Si semiconducter is kept at room temperature.
The thermal energy $= kT$

$$= 1.38 \times 10^{-23} \times 300$$

$$\approx 4.2 \times 10^{-21}$$

in terms of eV, $\approx \dfrac{4.2 \times 10^{-21}}{1.6 \times 10^{-19}}$ eV

$$\approx \boxed{8.617 \times 10^{-5}} \times 300$$

8.617
× 3
25851

$$= 25.851 \times 10^{-3}$$

CONCEPT OF EFFECTIVE MASS

→ $E = \frac{\hbar^2 k^2}{2m^*}$ ← m^* is now the eff. mass inside the SM.

⇒ $m^* = \frac{\hbar^2}{\frac{\partial^2 E}{\partial k^2}}$.

⇒ $m^* \propto \frac{1}{\left(\frac{\partial^2 E}{\partial k^2}\right)}$ → taken into account band structure info on the material

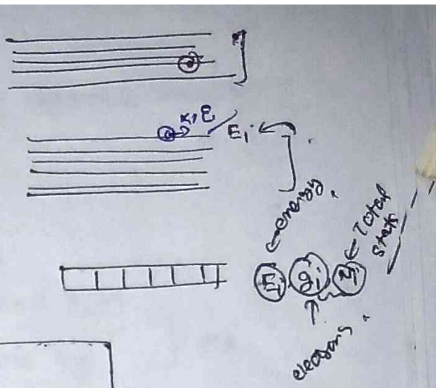
→ E-k is actually sort of band (structure)

** How momentum changes with energy

→ $\boxed{\text{All classical physical equations can be applied after taking } m^* \text{ like } F = ma}$ **

FERMI - DISTRIBUTION

probabity = $\frac{N_i}{g_i}$ → an energy state that can be occupied by an $e^{\ominus}$ at E_i leel

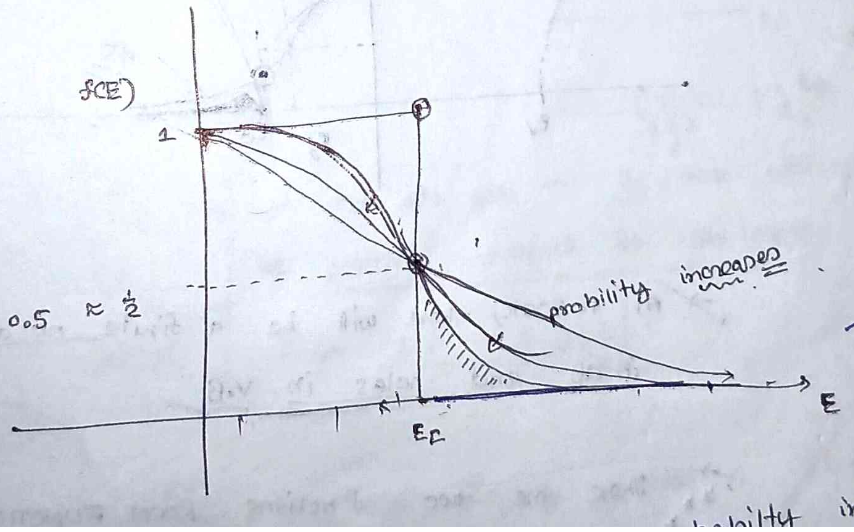


give this energy by this fermi function

$$\boxed{f(E_i) = \frac{N_i}{g_i} = \frac{1}{1 + \exp\left[\frac{E_i - E_f}{K_B T}\right]}}$$

E_f = fermi level → it is an statistical const/concept
→ energy level where finding $e^{\ominus}$ is 50%
→ it is independent of temperature

$f(E_f) = \frac{1}{2}$ [50%]



probility increases

Let's consider the energy of the electron is less than E_F and the considered is 0K

$$f(E) = \frac{1}{1+e^{-\infty}} \Rightarrow 1 .$$

Let's consider the energy of the electron is greater than E_F and the temperature is 0 K.

$$f(E) = \frac{1}{1+e^{+\infty}} \Rightarrow 0$$

At T = 0K, all the states below femi - energy level will be 1 and above the E_{femi} will be 0.

$e^{\ominus}$ con con tram $n = N_C e^{-\frac{(E_c - E_f)}{KT}}$

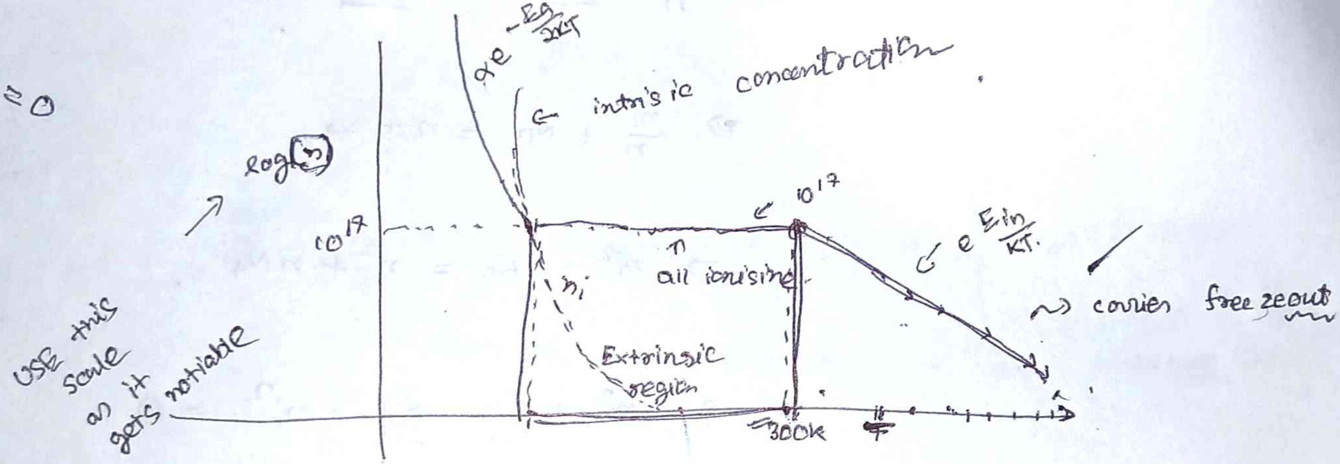
↳ eff density of sate / wells at energy level E_C.

hole concention $= n_p = N_V e^{-\frac{(E_f - E_V)}{KT}}$

↳ eff. density of state/wells at enrgy level E_V.

TEMPERATURE DEPENDENCE

On lowering, not all donors may not get ionised
$\Rightarrow$ free $e^{\ominus}$s conc reduces as well. and this is called carrier freeze-out.



graph explanation
→ At low temp, there is very low Donor electrons available in energy-type blond. , so as you decrease the no of $e^{\ominus}$s also gets decreased.

→ At some temp, all the donors get ionised and thus the graph remains flat and is the extrinsic region

DENISTY OF STATES

(1) DOS represents the number of available energy states per unit volume per unit energy.

$$g(E) = \int_{E_1}^{E_2} N(E)\, dE$$

N(E) = density of sate

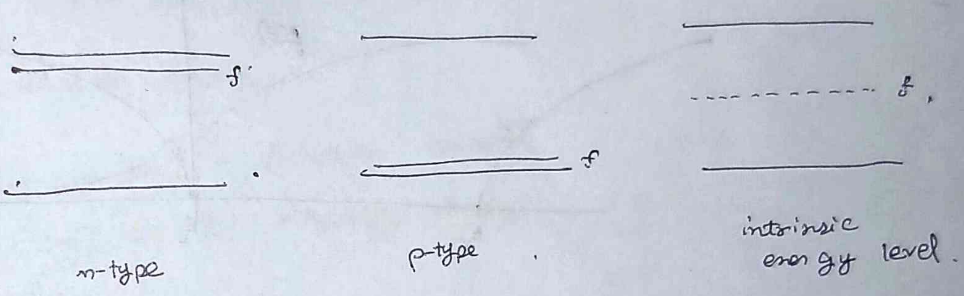
g(E) = no of allowed energy states per unit of volume

$$g(E) = \frac{4\pi}{h^3} (2m^*)^{3/2} \sqrt{(E - E_c)}$$, vaconcy

→ In an n-type semiconducter, the femi-level lies close to the conduction band.

→ in p type semiconducter, the femi level lies near the valance band.

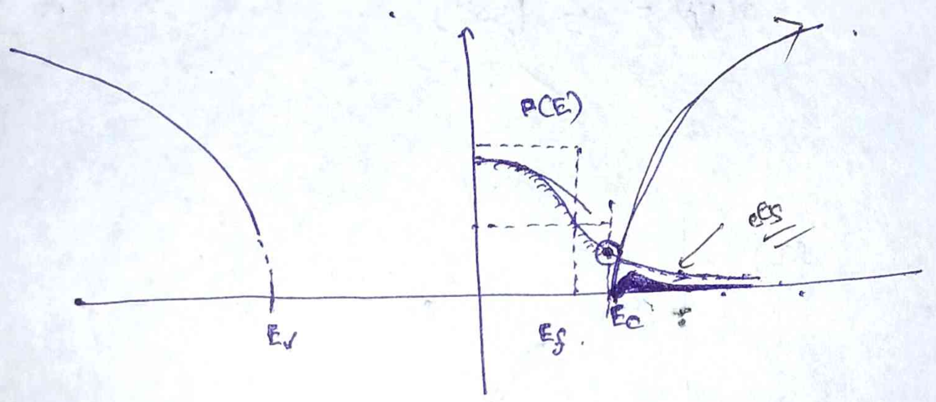
→ In an intrinsic semiconductor, the fermi level lies in the middle of the energy band gap



Total no of electrons in conduction band

how many states available

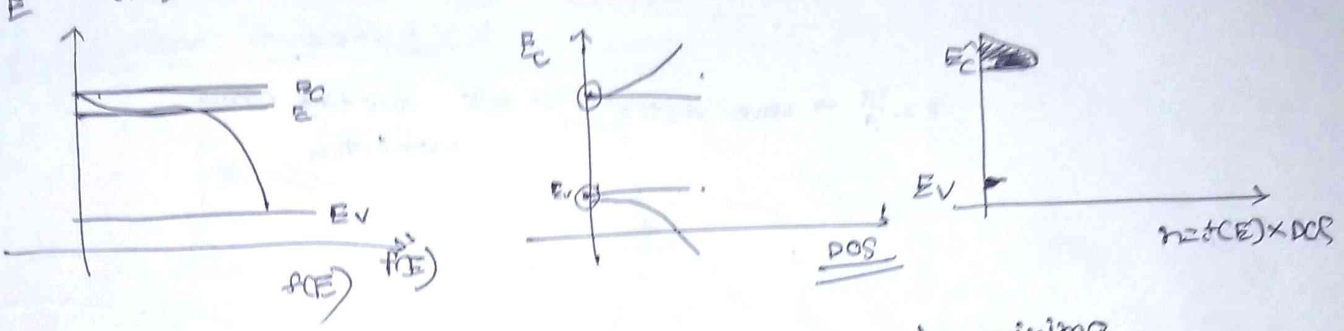
$$n = \int_{E_c}^{\infty} g(E)\, f(E)\, dE = \left(\text{how many states available}\right) \times \left(\text{probability that they occupy}\right) \times$$



At T= 300K, there will be a finite no of electrons in CB and holes in V.B

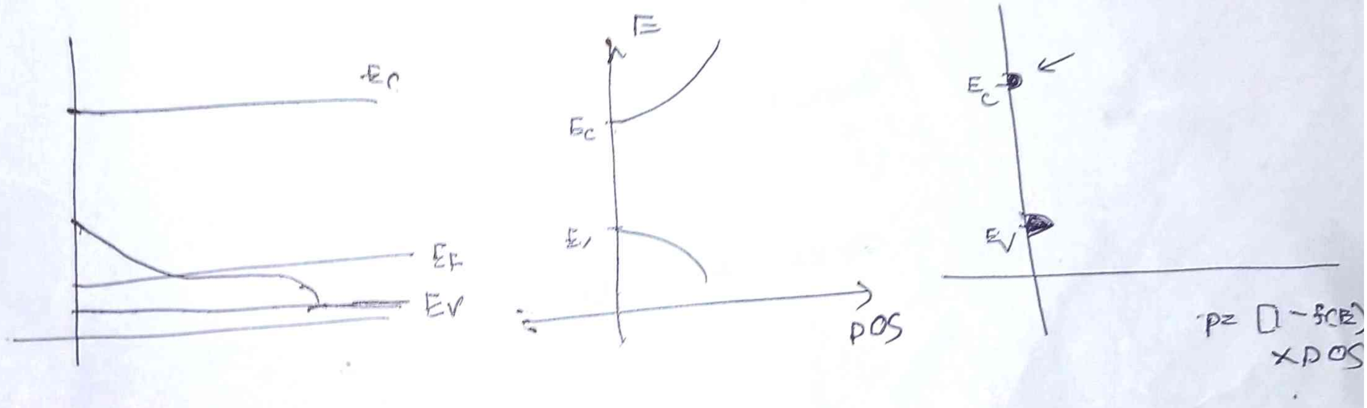
⇒ Total CARRIER ~~electron~~ concentration emission .

N type



electrons stay at the edge conduction band minima

P type

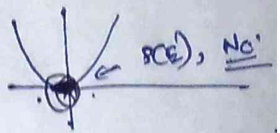


KT = 0.

Self explain

(i) Nc is effective conduction band D.O.S (cm^{-3}).

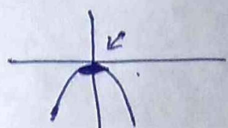
and it represents the no of rooms at E_c for elles to live and similar for holes



$$N_c = \left(\frac{2\pi m^* KT}{h^2}\right)^{3/2} \times$$

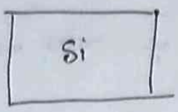
(ii)

$$N_v = \left[\frac{2\pi m_h^* K_B T}{h^2}\right]^{3/2}$$

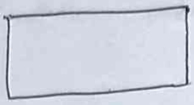


MASS ACTION LAW

→ The product of electron cocentration and hole concentration is constant at a given temperature and it is equal to square of intrinsic carrier concentration at that temp.



$n_i = 1.45 \times 10^{10} / cm^3$



→ Let the semiconductor is doped with acceptor concentration of 10^{15} cm^3

$N_a = 10^{15} / cm^3$

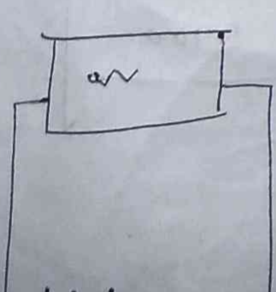
$n_i = 1.45 \times 10^{10}$

$p10^{15} \times n = [1.45 \times 10^{10}]^2$

$n = (1.45)^2 \times 10^5$ cm^3

→ The concentration is decreasing as there is recombination is taking place

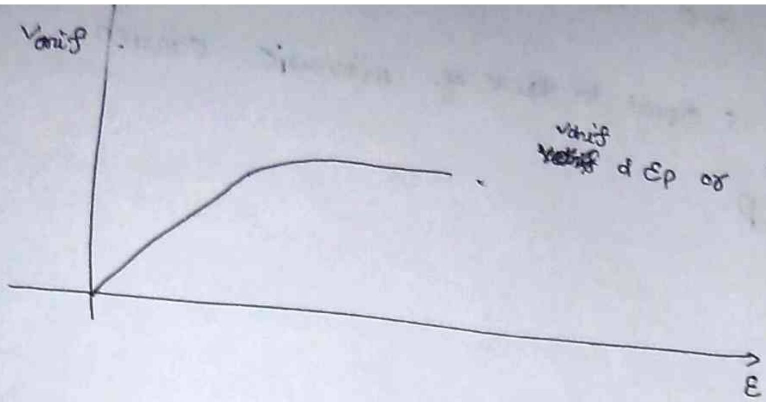
Drift of carriers in a semiconductor,



Let the semiconductor has hole concentration of p and electron concentration of n.

The current density due to hole may be written as :

$J = qp V_{diff-p}$



$V_{drif} \propto \mathcal{E}_p$ or $\underline{V_{drif} = \mu \mathcal{E}}$;

So, the drift velocity of holes may be written as

$$V_{drif} = \mu \mathcal{E}.$$

So, current density ~~for hole~~ due to may be written as

$$J_p = q\,p\,\mu_p \mathcal{E}_p.$$

Similarly, the total current density may be written as

$$J = J_p + J_n$$

$$= q p \mu_p \mathcal{E} + q n \mu_n \mathcal{E}$$

$$\boxed{J = q(p\mu_p + n\mu_n)\,\mathcal{E}}$$

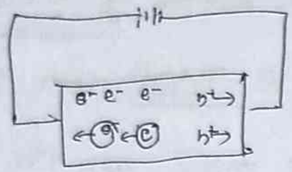
$$J = \sigma \mathcal{E}.$$

	Hole Mobility	Electron mobility
Si	450	1350
GaAs	400	8000
Ge	1900	3900

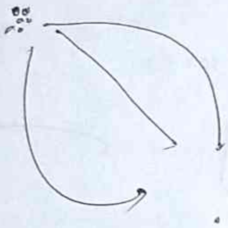
Drift of carriers in a semiconductor.

$$J_{drift} = [qN\mu_n + qp\mu_p]E$$

n-type



Diffusion of CARRIES IN SEMICONDUCTOR



p-type
→ Non-homogenous distribution

Homogenous distribution of carriers.

Flux

$$J_{diff} = qD_n \frac{dn}{dx}.$$

$J_{diffusion\_n}$ = ~~drift~~ diff... current due to electrons

$\frac{dn}{dx}$ = concentration ... electron...

D_n = diffusion coeffient

$$J_{diffusion-p} = -q D_p \frac{dp}{dx}$$

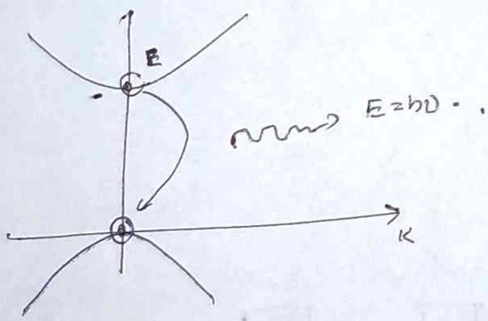
$\frac{dp}{dx}$ = concentration gradient of holes.

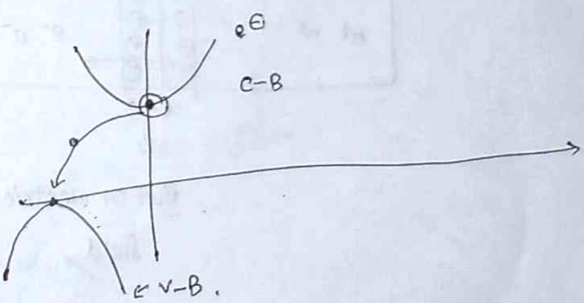
D_p = diffusion coeff for holes

Recombination of carriers.

→ Recombination of carriers may take in two processess.

① Direct recombination
② Indirect recombination.





↑ Some energy is lost as heat and other is emitted as energy

$\boxed{\text{direct band-gap semiconducter}}$

GaN
GaAs
GaP

Si
Ge
AlAs.

Carrier lifetime.

→ Low carrier lifetime

↓

DIODES

High carrier lifetime

↓

OTHERS.

— END OF UNIT 1 —